Name
Date
Course

Review and Expanding on Various Topics (Homework)

Determine if each of the following equations has one solution, no solution, or infinite solutions.
Demonstrate algebraically.

Answers

① $7x+2=16x-4-9x$ — No Solution

② $3x-5x=4+2x-1$ — One Solution

③ $-10x+6x-7=2(4+7x)$ — One Solution

④ $3x+2-1=-x+4x$ — No Solution

⑤ $-2(x-4)=5-2x+3$ — Infinite Solutions

⑥ $4=2-3x$ — One Solution

⑦ $3(3x-2)+4=-8+4(-2x+1)+17x+2$ — Infinite Solutions

⑧ $-9x-x=4x-14x$ — Infinite Solutions

⑨ $1-2(4-8x)=15x$ — One Solution

⑩ $-5-2+x=2(3x-5)-5x$ — No Solution

Determine if each of the following systems has one solution, no solution, or infinite solutions.
Demonstrate algebraically.

⑪ $\begin{cases} x+1=y \\ x+y=8 \end{cases}$ — One Solution

⑫ $\begin{cases} 2x-2y=1 \\ 16x-16y=8 \end{cases}$ — Infinite Solutions

⑬ $\begin{cases} -2x+y=14 \\ x=-3y \end{cases}$ — One Solution

⑭ $\begin{cases} -12x-8y=-4 \\ 3x+2y=1 \end{cases}$ — Infinite Solutions

⑮ $\begin{cases} -4x+3y=6 \\ \frac{4x}{3}+5=y \end{cases}$ — No Solution

⑯ $\begin{cases} 2x-2y=0 \\ y=x \end{cases}$ — Infinite Solutions

⑰ $\begin{cases} 5x+2y=-14 \\ 2x+3y=-10 \end{cases}$ — One Solution

⑱ $\begin{cases} x=4+2y \\ x-2y=-4 \end{cases}$ — No Solution

⑲ $\begin{cases} 2+4y=x \\ x=4y+1 \end{cases}$ — No Solution

* Watch your Signs.

⑳ $\begin{cases} y=3-2x \\ 2x-4y=7 \end{cases}$ — One Solution

Show the solutions of each system of inequalities by graphing.

Answers are on a different page.

㉑ $\begin{cases} y\geq 4x-2 \\ \frac{2}{3}(x+1)\geq y+1 \end{cases}$

㉒ $\begin{cases} 2x+3y>2 \\ 3x-2y<3 \end{cases}$

Continue on the next page.

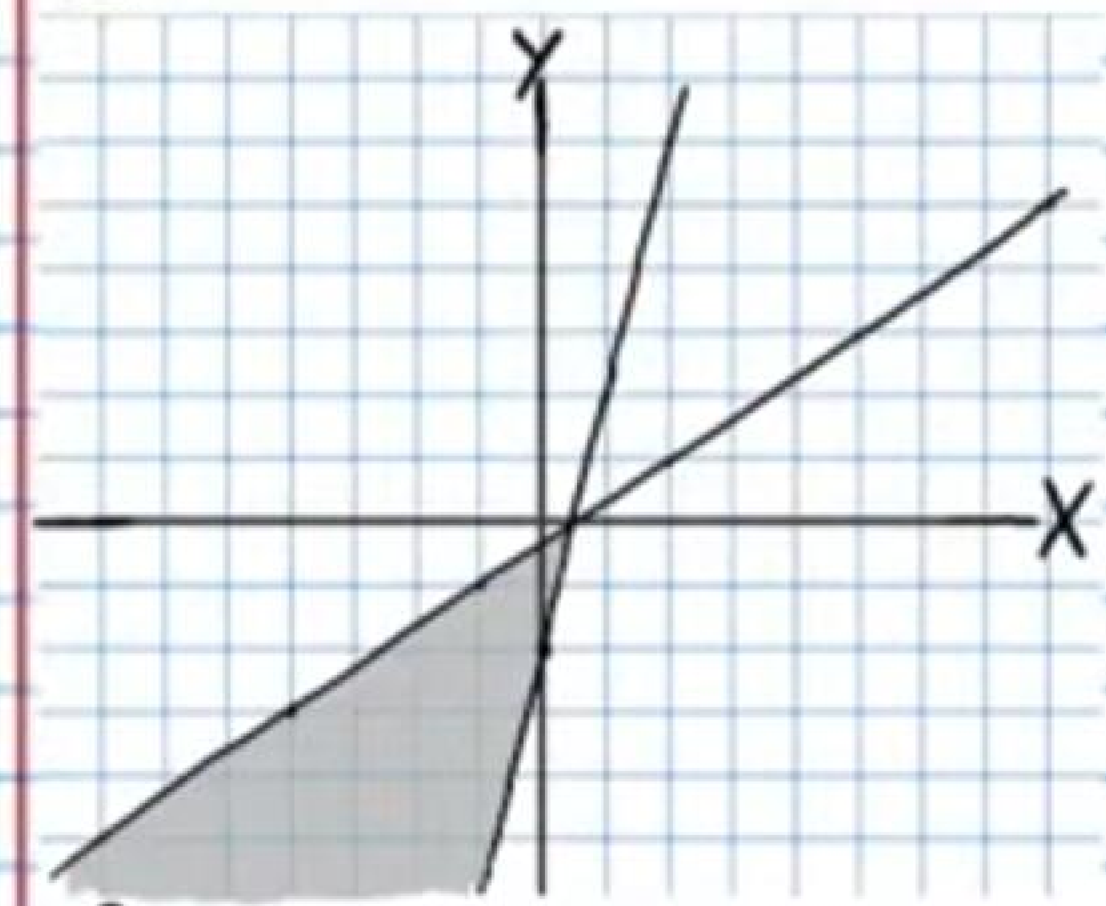
(23) $\begin{cases} -\frac{2}{5}x+2 \le y \\ y \ge x+4 \end{cases}$

(24) $\begin{cases} y-2 < 4(x-1) \\ y+3 \ge -\frac{1}{2}(x-3) \end{cases}$

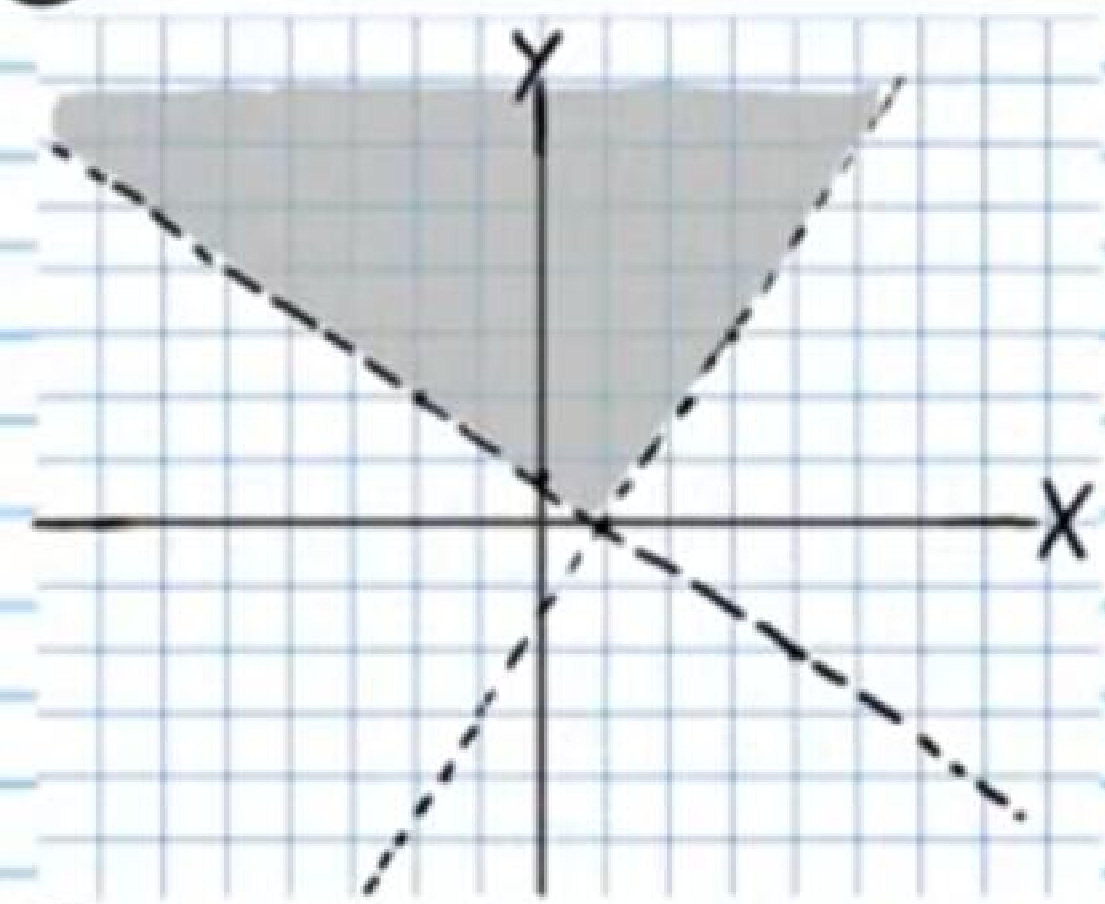
(25) $\begin{cases} y > -3x \\ -1 \ge x+2y \end{cases}$

Answers for Problem #'s 21-25

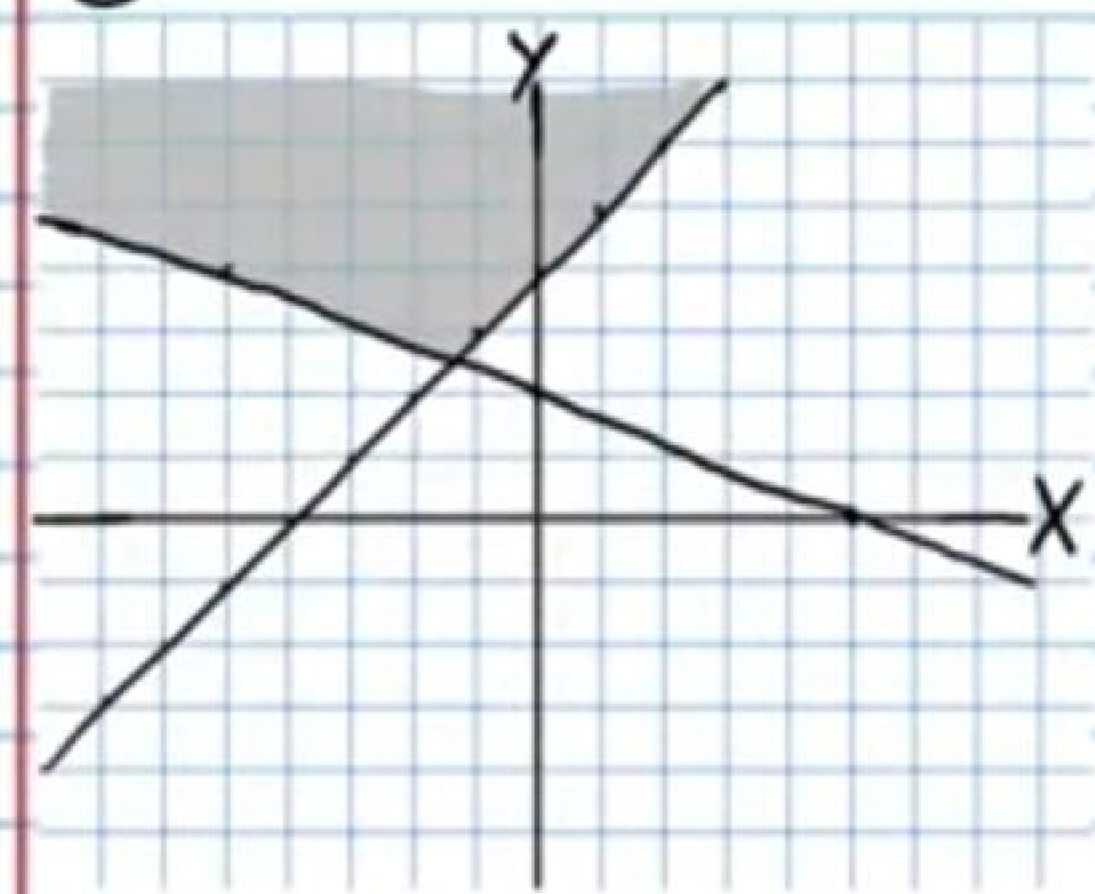
(21)



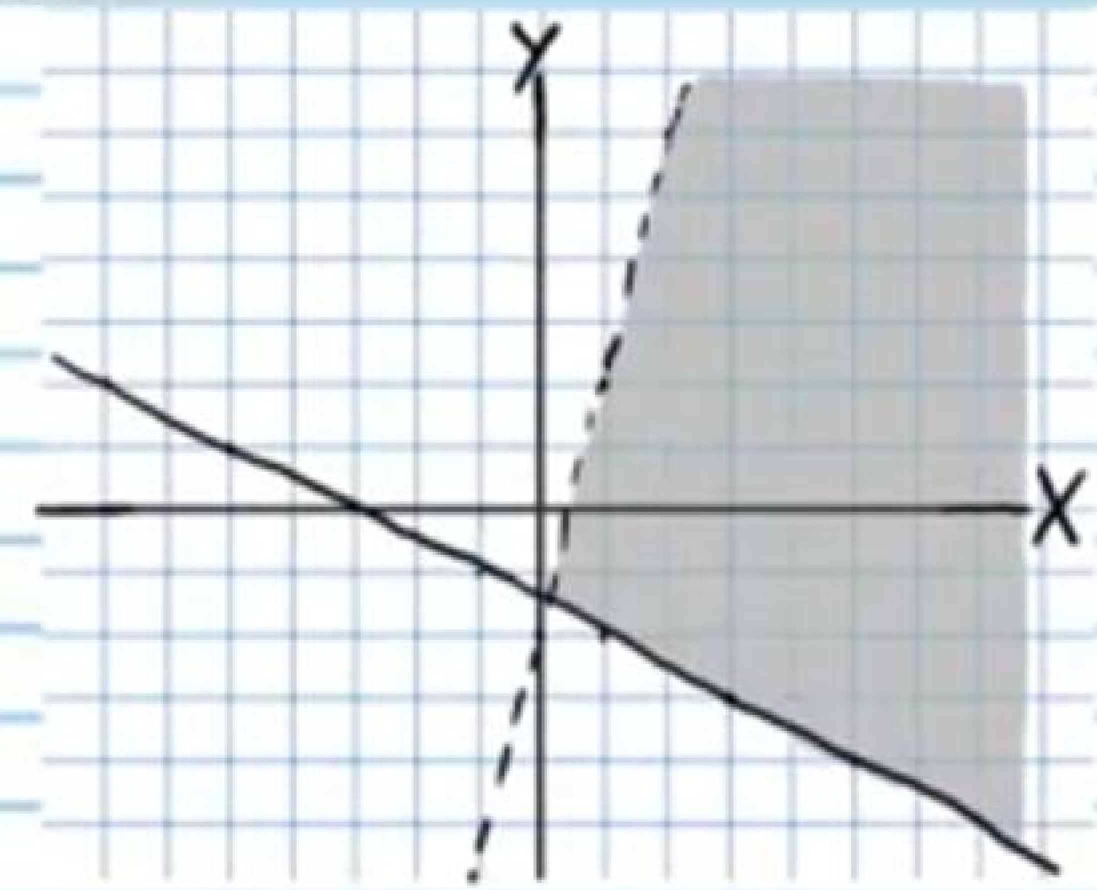
(22)



(23)



(24)



(25)